Sr.No. 8641

Bharatiya Vidya Bhavan's

SARDAR PATEL INSTITUTE OF TECHONOLOGY

(An Autonomous Institute Affiliated to University of Mumbai)

MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058

MID SEMESTER EXAMINATION

AY: 2025-26	Session: Odd MSE October 2025	Date: 08-10-2025
Sem: 1	Slot: 10:00am - 11:00am	C. Code: CS413
Exam Type: Written	Block:605 Course Section:1	Seat No: 2022300124

1_25_123_986_1_1505_41816

Name of the candidate. Sanika Sule

Candidate's UID number: 2 0 2 2 3 0 0 1 2 4

Examination class room number: 5 0 8

Invigilator's signature with date: 8/10/25

(To be filled in by the candidate)

EXAMINATION: BE TECH

(For example FY MCA/FY BTECH)

PROGRAM: Comps B

DAY AND DATE: Wednesday, 8th Oct 2025

COURSE NAME: DL.

SEMESTER : I / II / III / IV / V / VI / VII / VIII

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

KE/2025/03

1838

INSTRUCTIONS TO CANDIDATES

1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.

2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.

3) Do not write anything in the column provided for the marks to be assigned by the examiners.

4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.

5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.

6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.

7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.

8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting.**

9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.

10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.

11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.

12) The answer-books will be scrutinized before they are sent to examiners.

13) All answer-books supplied shall be returned whether written or blank.

14) Nothing shall be written on the question-paper.

15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.

16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.

17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Q. 1A Solution:-

Given,

$$f(\omega) = \frac{1}{2}\|X\omega - y\|^2 = \frac{1}{2}(X\omega - y)^2$$

~~Required gradient vector~~

$$\frac{\partial f(\omega)}{\partial x\omega} = \frac{1}{2} \times 2 \|X\omega - y\| \cdot X$$

$$\therefore \frac{\partial f(\omega)}{\partial \omega} = X \cdot (X\omega - y) = X^2\omega - Xy$$

$$\frac{\partial f(\omega)}{\partial y} = \frac{1}{2} \times 2 (X\omega - y)(-1)$$

$$\frac{\partial f(\omega)}{\partial y} = -X\omega + y.$$

$\therefore$ required gradient vector $= \Delta f(\omega)$

$$= \begin{bmatrix} \frac{\partial f(\omega)}{\partial \omega} \\ \frac{\partial f(\omega)}{\partial y} \end{bmatrix} = \begin{bmatrix} X^2\omega - Xy \\ -X\omega + y \end{bmatrix}$$

We know that,
Gradient update rule is given by

$$\omega_{k+1} = \omega_k - \eta \Delta f(\omega)$$

where
η = convergence rate.

$\therefore$ required gradient update rule

$$= \boxed{\omega_{k+1} = \omega_k - \eta \begin{bmatrix} X^2\omega - Xy \\ -X\omega + y \end{bmatrix}}$$

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also

$\frac{\partial f(\omega)}{\partial \omega^2} = x^2$, $\frac{\partial f(\omega)}{\partial y^2} = 1$

$\frac{\partial f(\omega)}{\partial y \partial \omega} = -x$, $\frac{\partial f(\omega)}{\partial \omega \partial y} = -x$

$\therefore$ Required Heissen matrix

$H = \begin{bmatrix} x^2 & -x \\ 1 & -x \end{bmatrix}$

so,

$\det(H - \lambda I) = 0$

$\det\left(\begin{bmatrix} x^2 & -x \\ 1 & -x \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) = 0$

$\det\left(\begin{bmatrix} x^2 - \lambda & -x \\ 1 & +x\lambda \end{bmatrix}\right) = 0$

$\therefore (x^2 - \lambda)(x\lambda) - (-x)(1) = 0$

$\therefore x^3\lambda - x\lambda^2 + x = 0$

$-x\lambda^2 + x^3\lambda + x = 0$

$x\lambda^2 \cancel{+} x^3\lambda - x = 0$

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Q.1b) Solution:-

1) Since model capacity is too high, the model tends to summarize entire dataset as it is instead of learning the patterns from training dataset.
2) This results in overfitting of the model.
3) ∵ the model is uncapable of learning patterns, prediction of testing dataset with a different parameter value would be incorrect.
4) This thereby affects generalization.
5) which means model is incapable of predicting correct ~~table~~ label / value for a testing point if one of its features values differ from that of ~~th~~ its correct label / value despite of belonging to same class.
6) For example, if the training dataset, includes features to identify a ball or not a ball, the features are learned as its is (like shape, color, texture);
7) If any object which is a ball but has slightly different feature value may ~~be~~ ~~dae~~ not be labelled as ball despite being a ball affecting the accuracy of the model.

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	Q.2a	Solution :-

1) For ~~d depth~~ standard 2D convolution,
let
$H \times W \times C$:= spatial dimension of input feature map
Cin := no. of input channels
Cout := no. of output channels.
$F \times F$:= Spatial dimension of filter matrix.
k := no. of filters.

2) For every output channel, a filter of size $F \times F \times Cin$ slides across the input feature map.

2) Since, there are K filters, ~~the~~ thus a total size $K \times F \times F \times Cin$ slide across the input feature map.

3) For every pos. where K filters are applied
Since there are approximately $H \times W \times C$ positions
∴ count of applied features is
$H \times W \times C \times K \times F \times F \times Cin$.

4) Thus for total Cout output channels, the total no of operations thus become
$H \times W \times C \times k \times F \times F \times Cin \times Cout$

5) Thus the computational complexity is
$\boxed{O(H \times W \times C \times K \times F \times F \times Cin \times Cout)}$.

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